

ARP

- Task 3

Recalling from previous tasks that devices can have two identifiers: A MAC address and IP Address, the Address Resolution protocol or ARP for short, is the technology that is responsible for allowing devices to identify themselves on network. *ARP is responsible for finding MAC (hardware) address related to specific IP address. It works by broadcasting an ARP query, "who has this IP address?"

Tell me." And the response is of the form, "The IP Address is at this MAC address".

*Simply ARP Allows a device to associate its MAC address with an IP address on network. Each device on network will keep log of MAC addresses associated with other devices.

*when devices wish to communicate with another, they will send a broadcast to the entire network searching for specific device. Devices can use ARP to find MAC Address of devices for communication

*~~Problem~~ The problem:- while we use IP addresses to route data across the internet, local network hardware (like Switches) doesn't understand IPs - they only speak MAC Addresses. If Device A knows Device B IP's but not its MAC, it can't actually deliver the data packet.

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How does ARP work?

Each device within a network has ledger to store information on, which is called cache.

In the context of ARP, this cache stores the identifiers of other devices on network.

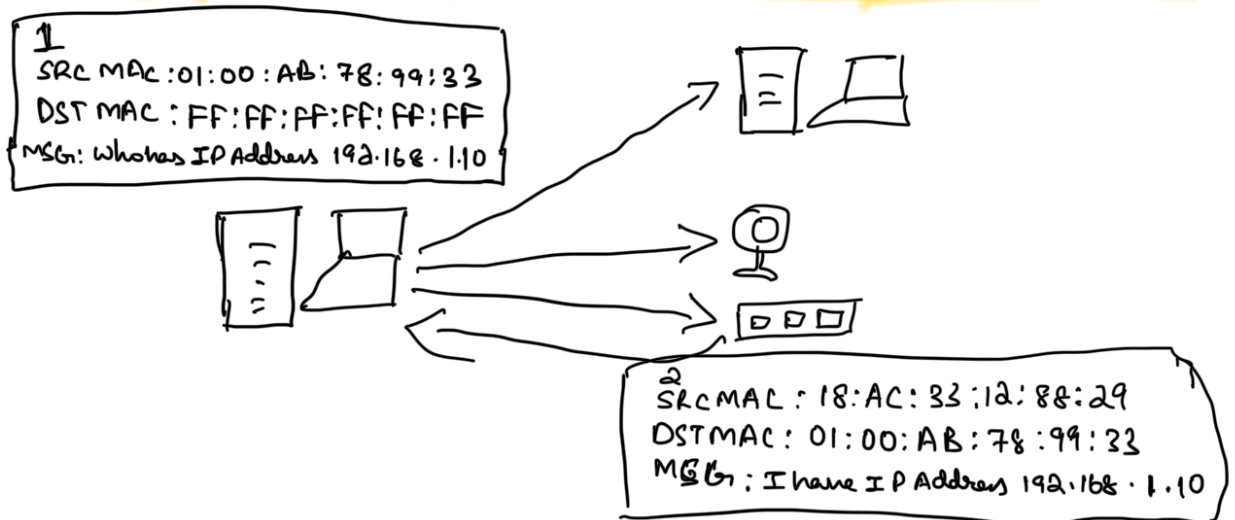
*In order to map these two identifiers together (IP address and MAC address), ARP sends two types of message

~~***~~ 1) ARP Request 2) ARP Reply

-> when ARP request is sent Device A sends a message in an ARP broadcast

to every other devices on local network, who has the IP Address 192.168.1.10?
please tell me your MAC address!

→ ARP reply : — the devices that actually owns that IP Address responds directly to device A : "That's me! Here is my MAC address: 00:AA:11:BB:22:CC"



1) What does ARP stands for?

A) Address Resolution Protocol

2) What category of ARP packet asks a device wheather or not it has a specific IP address?

A) Request

3) What address is used as a physical identifier for a device on network?

A) MAC Address

4) What address is used as a logical identifier for a device on network?

A) IP Address.